TEACHER GUIDE

Pearson Edexcel International Advanced Level in Business (XBS11 / YBS11)

Investment appraisal

Unit 3: Business decisions and strategy
3.3.3 Decision-making techniques: investment appraisal



A revision lesson. Pupils run a car dealership in firms of four and revise investment appraisal on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same.

1 Average profit

Round 1

New charging points.

Average profit each year?

Cost to set up	**£30,000**
Returns over its life	**£66,000**
It lasts	**4 years**

ANSWER

£9,000

(£66,000 − £30,000) ÷ 4 = £9,000

Why: Average yearly profit is total profit shared over the years.

2 Limits of payback

Round 1

You choose between two projects by payback.

Which two are limits of payback?

A It ignores the time value of money
B It favours slow projects
C It ignores the project's cost
D It ignores returns after payback
E It is hard to work out

ANSWER

✓ It ignores the time value of money
✓ It ignores returns after payback

Why: Payback stops at the payback point and treats a pound later as a pound now.

3 Rank by payback

Round 2

Three projects with even yearly returns.

Order by payback, fastest first?

A Cost £40,000, £10,000 a year
B Cost £30,000, £12,000 a year
C Cost £60,000, £20,000 a year

ANSWER

1. **Cost £30,000, £12,000 a year**
2. **Cost £60,000, £20,000 a year**
3. **Cost £40,000, £10,000 a year**

Why: Payback: 2.5, 3 and 4 years.

Topic tasks

4 Payback in months

Round 2

A new valeting bay.

Payback period, in months?

Cost to set up	**£17,100**
Returns, each year	**£8,290**
Running costs, each year	**£1,450**

ANSWER

30 months

£8,290 − £1,450 = £6,840;
£17,100 ÷ £6,840 × 12 = 30

Why: It has paid for itself after 30 months.

5 Ramp or bay

Round 3

Ramp: payback 2 years, ARR 10%. Bay: payback 4 years, ARR 18%. Demand is stable; cash is fine.

Which project should you choose, and why?

A The ramp

B The bay

ANSWER

✔ **The bay**

Why: With little risk and cash to spare, the higher return wins.

6 Quicker payback

Round 3

Which pays back first?

Payback of the quicker option, in months?

Option A: cost	**£17,700**
Option A: returns a year	**£7,080**
Option B: cost	**£38,880**
Option B: returns a year	**£12,960**

ANSWER

£30

A 30 months; B 36 months; quicker 30

Why: The shorter payback gets the money back sooner.

Final questions

Each pupil answers 6 final questions on investment appraisal. They come from these 13 questions.

1. An inspection line costs £300,000 and should make £45,000 average annual profit. On original cost, what is its accounting rate of return?
   ✔ **15%**

2. A £180,000 paint booth returns £60,000, £75,000 and £90,000 over three years. Ignoring timing within years, what is its payback period?
   ✔ **2.5 years**

3. A charging hub has a 0.65 probability of producing a £240,000 payoff and a 0.35 probability of a £60,000 payoff. What is the expected payoff?
   ✔ **£177,000**

4. A project requires £120,000 now and has discounted future cash inflows of £138,500. What is its net present value?
   ✔ **£18,500**

5. A project costs £200,000 and its discounted future cash inflows total £224,000. What is NPV?
   ✔ **£24,000**

6. A system costs £250,000 and is expected to earn average annual profit of £30,000. What is ARR using original cost?
   ✔ **12%**

7. A project yields £300,000 with probability 0.4 or £80,000 with probability 0.6. It costs £130,000. What is expected net monetary value?
   ✔ **£38,000**

8. An energy-saving project costs £180,000. Discounted savings are £205,000, but a downside scenario reduces those savings by 15%. What is NPV in the downside scenario?
   ✔ **−£5,750**

9. A £90,000 paint booth brings in £30,000 in year 1, then £40,000 a year. What is its payback period?
   ✔ **2.5 years**

10. A dealer spends £60,000 on a new workshop ramp, which brings a net cash inflow of £24,000 a year. What is the payback period?
    ✔ **2.5 years**

Final questions

11 Two projects cost the same. Project A pays back in 2 years and project B in 4 years. Why might a cautious dealer prefer A?

✔ **It gets its money back sooner, so there is less risk**

12 A dealer invests £80,000 in a valeting centre. Over 5 years the total profit, after taking off the £80,000 cost, is £40,000. What is the ARR?

✔ **10%**

13 Project X has an ARR of 14% and project Y an ARR of 9%. The dealer borrows at 11% a year. On ARR alone, what follows?

✔ **Only project X earns more than the cost of borrowing**